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Институт №8 “Компьютерные науки и прикладная математика” Кафедра №806 “Вычислительная математика и программирование”

Лабораторная работа №2 по курсу

«Операционные системы»

Группа: М8О-211Б-23

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Преподаватель: Бахарев В.Д.

Оценка: _______

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Постановка задачи

Цель работы:

Целью является приобретение практических навыков в управление потоками в ОС и обеспечение синхронизации между потоками.

Задание:

Составить программу на языке Си, обрабатывающую данные в многопоточном режиме. При обработки использовать стандартные средства создания потоков операционной системы (Windows/Unix). Ограничение максимального количества потоков, работающих в один момент времени, должно быть задано ключом запуска вашей программы. Так же необходимо уметь продемонстрировать количество потоков, используемое вашей программой с помощью стандартных средств операционной системы. В отчете привести исследование зависимости ускорения и эффективности алгоритма от входных данных и количества потоков. Получившиеся результаты необходимо объяснить.

Вариант 4.

Отсортировать массив целых чисел при помощи сортировки слиянием.

Общий метод и алгоритм решения

Использованные системные вызовы:

- int pthread_create(pthread_t *__restrict__newthread, const pthread_attr_t * restrict__attr, void *(*__start_routine)(void *), void *__restrict__arg) — создаёт поток с рутиной (стартовой функцией) и заданными аргументами
- int pthread_join(pthread_t __th, void **_thread_return) — дожидается завершения потока (Эта процедура блокирует вызывающий поток до тех пор, пока не будет

осуществлен выход из другого (указанного) потока).

Для разграничения доступа потоков к общим ресурсам могут используем мьютекс. Мьютекс представляет объект, который может находиться в двух состояниях: заблокированном и разблокированном. На уровне языка Си мьютекс представлен типом pthread_mutex_t.

int pthread_mutex_init(pthread_mutex_t *mutex, const pthread_mutexattr_t *mutexattr) – инициализация мьютекса;

int pthread_mutex_lock(pthread_mutex_t *mutex) – блокировка мьютекса;

int pthread_mutex_unlock(pthread_mutex_t *mutex) – разблокировка мьютекса;

int pthread_mutex_destroy(pthread_mutex_t *mutex) – уничтожение мьютекса;

Программа получает на вход два аргумента – размер массива и максимально количество потоков для сортировки. Для полученного размера массива, происходит его заполнение случайными значениями с помощью функции srand().

Мьютекс используется для синхронизации доступа к глобальной переменной thread_count, отслеживающей общее количество созданных потоков.

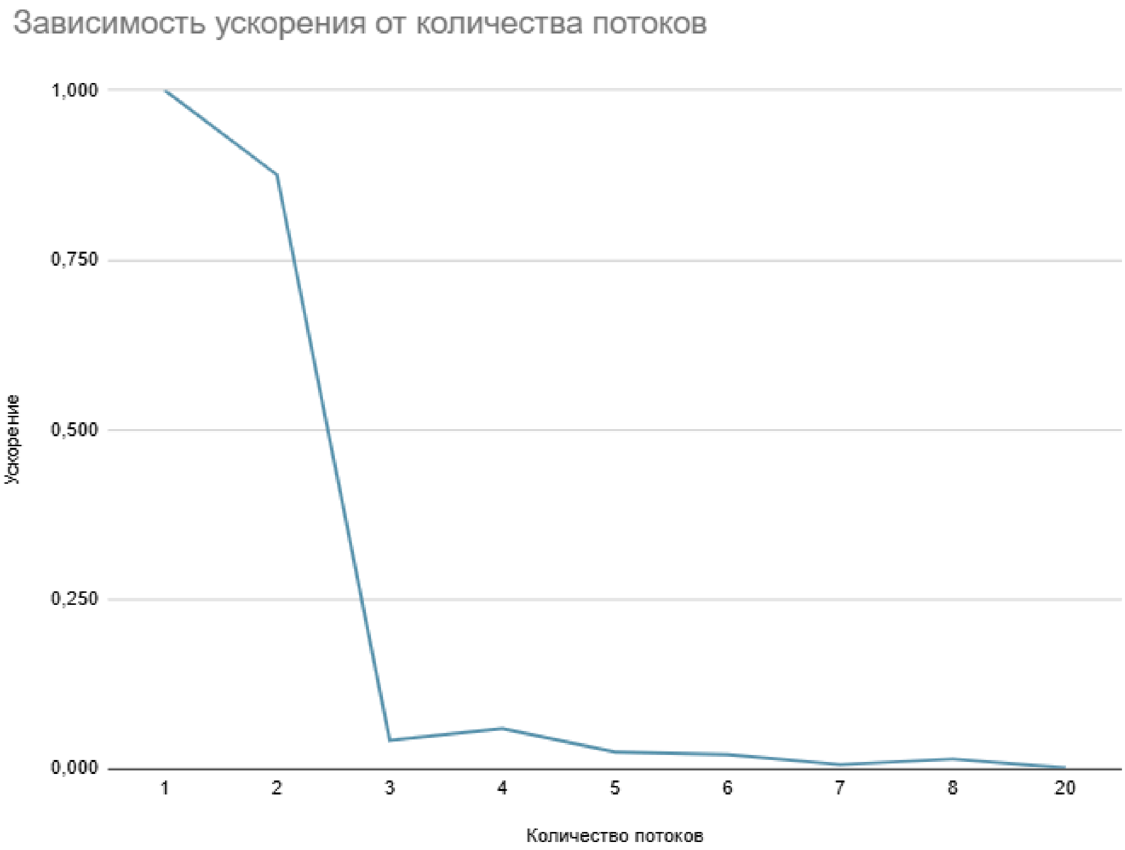
Сортировка слиянием

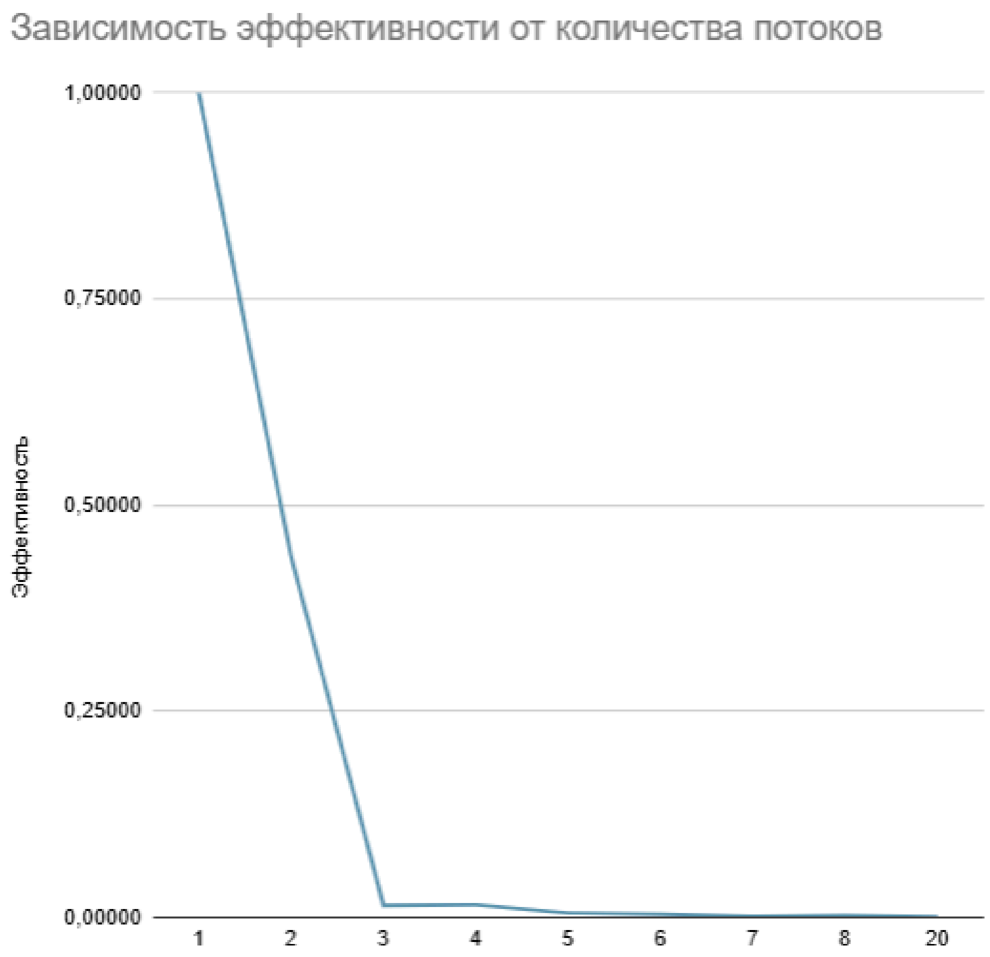
Разбиваем текущий массив пополам, сливаем левый и правый подмассив. По правилу: если элемент из левого подмассива меньше элемента из правого подмассива записываем левый в результирующий массив и увеличиваем индекс, отвечающий за начало левого массива, иначе записываем элемент из правого подмассива и сдвигаем указатель, указывающий на начало правого массива. Сортировку производим рекурсивно. При рекурсии создаем новые потоки, чтобы левая и правая части массива сортировались параллельно.

Ниже приведены данные, показывающие изменение ускорения и эффективности, с разным количеством потоков, для этой реализации. Размер массива 1000.

Число потоков	Время выполнения (с)	Ускорение	Эффективность
1	0,000007	1,000	1,00000
2	0,000008	0,875	0,43750
3	0,000167	0,042	0,01397
4	0,000118	0,059	0,01483
5	0,000282	0,025	0,00496
6	0,000332	0,021	0,00351
7	0,001129	0,006	0,00089
8	0,000481	0,015	0,00182
20	0,004878	0,001	0,00007

Размер массива	Время выполнения(с)
2	0.000002
4	0.000001
100	0.000002
1000	0.000007
15000	0.000093
150001	0.009052
100000	0.000621
1000000	0.027520





Зависимость эффективности от количества потоков
Эффективность
1,00000
0,75000
0,50000
0,25000
0,00000
1
2
3
4
5
6
7
8
20

Код программы

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <time.h>

typedef struct
{
    int *array;
    int left;
    int right;
    int cnt;
} MergeSortArgs;

pthread_mutex_t mutex; // Мьютекс для синхронизации
int thread_count = 0;  // Количество активных потоков

void merge(int *array, int left, int mid, int right)
{
    int i, j, k;
    int n1 = mid - left + 1;
    int n2 = right - mid;

    int *L = (int *)malloc(n1 * sizeof(int));
    int *R = (int *)malloc(n2 * sizeof(int));

    for (i = 0; i < n1; i++)
        L[i] = array[left + i];
    for (j = 0; j < n2; j++)
```

```
        R[j] = array[mid + 1 + j];

    i = 0;
    j = 0;
    k = left;

    while (i < n1 && j < n2)
    {
        if (L[i] <= R[j])
            array[k++] = L[i++];
        else
            array[k++] = R[j++];
    }

    while (i < n1)
        array[k++] = L[i++];
    while (j < n2)
        array[k++] = R[j++];

    free(L);
    free(R);
}

void *merge_sort(void *args)
{
    MergeSortArgs *ms_args = (MergeSortArgs *)args;
    int left = ms_args->left;
    int right = ms_args->right;
    int mid;
    int cnt = ms_args->cnt;
```

```
pthread_mutex_lock(&mutex);
thread_count += 2; // Увеличение счетчика активных потоков
pthread_mutex_unlock(&mutex);

if (left < right)
{
    mid = left + (right - left) / 2;

    MergeSortArgs left_args = {ms_args->array, left, mid, cnt};
    MergeSortArgs right_args = {ms_args->array, mid + 1, right, cnt};

    pthread_t thread1, thread2;

    if (thread_count < cnt)
    {
        pthread_create(&thread1, NULL, merge_sort, &left_args);
        pthread_create(&thread2, NULL, merge_sort, &right_args);

        pthread_join(thread1, NULL);
        pthread_join(thread2, NULL);
    }

    merge(ms_args->array, left, mid, right);
}

pthread_mutex_lock(&mutex);
--thread_count; // Уменьшение счетчика активных потоков
pthread_mutex_unlock(&mutex);
```

```
    return NULL;
}

int main(int argc, char *argv[])
{
    if (argc != 3)
    {
        printf("Usage: %s <array_size> <num_threads>\n", argv[0]);
        return 1;
    }

    int n = atoi(argv[1]);
    int num_threads = atoi(argv[2]);

    int *arr = (int *)malloc(n * sizeof(int));
    srand(100);
    for (int i = 0; i < n; i++)
    {
        arr[i] = rand() % 100;
    }

    pthread_mutex_init(&mutex, NULL);
    MergeSortArgs args = {arr, 0, n - 1, num_threads};

    clock_t start_time = clock();
    merge_sort(&args);
    clock_t end_time = clock();

    double time_spent = (double)(end_time - start_time) / CLOCKS_PER_SEC;
    ++thread_count;
```

```c
    printf("The value of the requested threads: %d\n", num_threads);
    printf("The total number of threads created: %d\n", thread_count);
    printf("Elapsed time: %f seconds\n", time_spent);

    free(arr);
    pthread_mutex_destroy(&mutex);
    return 0;
}
```

Протокол работы программы

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 1

execve("./a.out", ["./a.out", "1000", "1"], 0x7fff739020f0 /* 30 vars */) = 0

brk(NULL) = 0x559f27345000

arch_prctl(0x3001 /* ARCH_??? */, 0x7ffe4ff8b320) = -1 EINVAL (Invalid argument)

mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5d5865d000

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0

mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f5d58656000

close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784

pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) = 48

pread64(3, "\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68, 896) = 68

newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0

pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) = 784

mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f5d5842d000

mprotect(0x7f5d58455000, 2023424, PROT_NONE) = 0

mmap(0x7f5d58455000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f5d58455000

```
mmap(0x7f5d585ea000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f5d585ea000
mmap(0x7f5d58643000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f5d58643000
mmap(0x7f5d58649000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f5d58649000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5d5842a000
arch_prctl(ARCH_SET_FS, 0x7f5d5842a740) = 0
set_tid_address(0x7f5d5842aa10)         = 11414
set_robust_list(0x7f5d5842aa20, 24)     = 0
rseq(0x7f5d5842b0e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f5d58643000, 16384, PROT_READ) = 0
mprotect(0x559f26671000, 4096, PROT_READ) = 0
mprotect(0x7f5d58697000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f5d58656000, 24583)           = 0
getrandom("\xf4\x93\xe6\x08\x99\x0a\x01\xed", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x559f27345000
brk(0x559f27366000)                     = 0x559f27366000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4914600}) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5033000}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 1
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 2
) = 39
```

```
write(1, "Elapsed time: 0.000119 seconds\n", 31Elapsed time: 0.000119 seconds
) = 31
exit_group(0)                     = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 2
execve("./a.out", ["./a.out", "1000", "2"], 0x7fff7aa31d40 /* 30 vars */) = 0
brk(NULL)                         = 0x55a46526d000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffea74a2910) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdd15873000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fdd1586c000
close(3)                          = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fdd15643000
```

mprotect(0x7fdd1566b000, 2023424, PROT_NONE) = 0
mmap(0x7fdd1566b000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7fdd1566b000
mmap(0x7fdd15800000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7fdd15800000
mmap(0x7fdd15859000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7fdd15859000
mmap(0x7fdd1585f000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fdd1585f000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdd15640000
arch_prctl(ARCH_SET_FS, 0x7fdd15640740) = 0
set_tid_address(0x7fdd15640a10) = 11450
set_robust_list(0x7fdd15640a20, 24) = 0
rseq(0x7fdd156410e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7fdd15859000, 16384, PROT_READ) = 0
mprotect(0x55a464d5e000, 4096, PROT_READ) = 0
mprotect(0x7fdd158ad000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7fdd1586c000, 24583) = 0
getrandom("\x90\x69\x77\x83\x25\xd1\x6b\x7b", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x55a46526d000
brk(0x55a46528e000) = 0x55a46528e000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4333200}) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4368600}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 2

```
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 2
) = 39
write(1, "Elapsed time: 0.000035 seconds\n", 31Elapsed time: 0.000035 seconds
) = 31
exit_group(0)                           = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 3
execve("./a.out", ["./a.out", "1000", "3"], 0x7ffd74c38790 /* 30 vars */) = 0
brk(NULL)                               = 0x56403b743000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc5cc9b9b0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f3a72770000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f3a72769000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
```

```
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f3a72540000
mprotect(0x7f3a72568000, 2023424, PROT_NONE) = 0
mmap(0x7f3a72568000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f3a72568000
mmap(0x7f3a726fd000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f3a726fd000
mmap(0x7f3a72756000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f3a72756000
mmap(0x7f3a7275c000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f3a7275c000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f3a7253d000
arch_prctl(ARCH_SET_FS, 0x7f3a7253d740) = 0
set_tid_address(0x7f3a7253da10)         = 11482
set_robust_list(0x7f3a7253da20, 24)     = 0
rseq(0x7f3a7253e0e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f3a72756000, 16384, PROT_READ) = 0
mprotect(0x56403a4ab000, 4096, PROT_READ) = 0
mprotect(0x7f3a727aa000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f3a72769000, 24583)           = 0
getrandom("\xcf\xd9\xfe\x55\x04\xab\x11\x60", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x56403b743000
brk(0x56403b764000)                     = 0x56403b764000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4228200}) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7f3a725d1870, sa_mask=[],
```

sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
sa_restorer=0x7f3a72582520}, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f3a71d3c000
mprotect(0x7f3a71d3d000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
**clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f3a7253c910,
parent_tid=0x7f3a7253c910, exit_signal=0, stack=0x7f3a71d3c000,
stack_size=0x7fff00, tls=0x7f3a7253c640} => {parent_tid=[11483]}, 88) = 11483**
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f3a7153b000
mprotect(0x7f3a7153c000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
**clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f3a71d3b910,
parent_tid=0x7f3a71d3b910, exit_signal=0, stack=0x7f3a7153b000,
stack_size=0x7fff00, tls=0x7f3a71d3b640} => {parent_tid=[11484]}, 88) = 11484**
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
**futex(0x7f3a7253c910, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME,
11483, NULL, FUTEX_BITSET_MATCH_ANY) = 0**
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4632400}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 3
) = 38

write(1, "The total number of threads crea"..., 39The total number of threads created: 4
) = 39
write(1, "Elapsed time: 0.000404 seconds\n", 31Elapsed time: 0.000404 seconds
) = 31
exit_group(0)   = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 4
execve("./a.out", ["./a.out", "1000", "4"], 0x7ffe646f38a0 /* 30 vars */) = 0
brk(NULL)   = 0x5606ce069000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc68a58b00) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fb65323d000
access("/etc/ld.so.preload", R_OK)   = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fb653236000
close(3)   = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784

```
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fb65300d000
mprotect(0x7fb653035000, 2023424, PROT_NONE) = 0
mmap(0x7fb653035000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7fb653035000
mmap(0x7fb6531ca000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7fb6531ca000
mmap(0x7fb653223000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7fb653223000
mmap(0x7fb653229000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fb653229000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fb65300a000
arch_prctl(ARCH_SET_FS, 0x7fb65300a740) = 0
set_tid_address(0x7fb65300aa10)         = 11490
set_robust_list(0x7fb65300aa20, 24)     = 0
rseq(0x7fb65300b0e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7fb653223000, 16384, PROT_READ) = 0
mprotect(0x5606ccc7d000, 4096, PROT_READ) = 0
mprotect(0x7fb653277000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7fb653236000, 24583)           = 0
getrandom("\x89\x96\x3c\xc6\x77\x4e\x16\x3f", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x5606ce069000
brk(0x5606ce08a000)                     = 0x5606ce08a000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4024200}) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7fb65309e870, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
```

sa_restorer=0x7fb65304f520}, NULL, 8) = 0

rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7fb652809000

mprotect(0x7fb65280a000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8)   = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fb653009910, parent_tid=0x7fb653009910, exit_signal=0, stack=0x7fb652809000, stack_size=0x7fff00, tls=0x7fb653009640} => {parent_tid=[11491]}, 88) = 11491

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7fb652008000

mprotect(0x7fb652009000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8)   = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fb652808910, parent_tid=0x7fb652808910, exit_signal=0, stack=0x7fb652008000, stack_size=0x7fff00, tls=0x7fb652808640} => {parent_tid=[11492]}, 88) = 11492

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

futex(0x7fb652808910, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 11492, NULL, FUTEX_BITSET_MATCH_ANY) = -1 EAGAIN (Resource temporarily unavailable)

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4556700}) = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...}, AT_EMPTY_PATH) = 0

write(1, "The value of the requested threa"..., 38The value of the requested threads: 4
) = 38

write(1, "The total number of threads crea"..., 39The total number of threads created: 4
) = 39
write(1, "Elapsed time: 0.000532 seconds\n", 31Elapsed time: 0.000532 seconds
) = 31
exit_group(0)   = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 5
execve("./a.out", ["./a.out", "1000", "5"], 0x7ffce49d5090 /* 30 vars */) = 0
brk(NULL)   = 0x55edea49a000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffdd2ceb5e0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f61eb073000
access("/etc/ld.so.preload", R_OK)   = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f61eb06c000
close(3)   = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784

```
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f61eae43000
mprotect(0x7f61eae6b000, 2023424, PROT_NONE) = 0
mmap(0x7f61eae6b000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f61eae6b000
mmap(0x7f61eb000000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f61eb000000
mmap(0x7f61eb059000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f61eb059000
mmap(0x7f61eb05f000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f61eb05f000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f61eae40000
arch_prctl(ARCH_SET_FS, 0x7f61eae40740) = 0
set_tid_address(0x7f61eae40a10)         = 11501
set_robust_list(0x7f61eae40a20, 24)     = 0
rseq(0x7f61eae410e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f61eb059000, 16384, PROT_READ) = 0
mprotect(0x55ede90f9000, 4096, PROT_READ) = 0
mprotect(0x7f61eb0ad000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f61eb06c000, 24583)           = 0
getrandom("\x33\x13\xf1\x31\xa0\xf5\xf8\x58", 8, GRND_NONBLOCK) = 8
brk(NULL)                               = 0x55edea49a000
brk(0x55edea4bb000)                     = 0x55edea4bb000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4062300}) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7f61eaed4870, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
```

sa_restorer=0x7f61eae85520}, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f61ea63f000
mprotect(0x7f61ea640000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f61eae3f910, parent_tid=0x7f61eae3f910, exit_signal=0, stack=0x7f61ea63f000, stack_size=0x7fff00, tls=0x7f61eae3f640} => {parent_tid=[11502]}, 88) = 11502
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f61e9e3d910, parent_tid=0x7f61e9e3d910, exit_signal=0, stack=0x7f61e963d000, stack_size=0x7fff00, tls=0x7f61e9e3d640} => {parent_tid=[0]}, 88) = 11505
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5301300}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 5
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 6
) = 39
write(1, "Elapsed time: 0.001239 seconds\n", 31Elapsed time: 0.001239 seconds
) = 31
exit_group(0) = ?
+++ exited with 0 +++

```
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 6
execve("./a.out", ["./a.out", "1000", "6"], 0x7fff603483b0 /* 30 vars */) = 0
brk(NULL)                               = 0x5563254d6000
arch_prctl(0x3001 /* ARCH_??? */, 0x7fff67916260) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f3567583000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f356757c000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f3567353000
mprotect(0x7f356737b000, 2023424, PROT_NONE) = 0
mmap(0x7f356737b000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f356737b000
mmap(0x7f3567510000, 360448, PROT_READ,
```

MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f3567510000
mmap(0x7f3567569000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f3567569000
mmap(0x7f356756f000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f356756f000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f3567350000
arch_prctl(ARCH_SET_FS, 0x7f3567350740) = 0
set_tid_address(0x7f3567350a10) = 11520
set_robust_list(0x7f3567350a20, 24) = 0
rseq(0x7f35673510e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f3567569000, 16384, PROT_READ) = 0
mprotect(0x556324c85000, 4096, PROT_READ) = 0
mprotect(0x7f35675bd000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f356757c000, 24583) = 0
getrandom("\x90\xf8\x3f\x68\xbb\xfa\x6b\x1d", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x5563254d6000
brk(0x5563254f7000) = 0x5563254f7000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4615600}) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7f35673e4870, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
sa_restorer=0x7f3567395520}, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f3566b4f000
mprotect(0x7f3566b50000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f356734f910, parent_tid=0x7f356734f910, exit_signal=0, stack=0x7f3566b4f000, stack_size=0x7fff00, tls=0x7f356734f640} => {parent_tid=[11521]}, 88) = 11521

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f356534c000

mprotect(0x7f356534d000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f3565b4c910, parent_tid=0x7f3565b4c910, exit_signal=0, stack=0x7f356534c000, stack_size=0x7fff00, tls=0x7f3565b4c640} => {parent_tid=[0]}, 88) = 11524

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5939300}) = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...}, AT_EMPTY_PATH) = 0

write(1, "The value of the requested threa"..., 38The value of the requested threads: 6

) = 38

write(1, "The total number of threads crea"..., 39The total number of threads created: 6

) = 39

write(1, "Elapsed time: 0.001324 seconds\n", 31Elapsed time: 0.001324 seconds

) = 31

exit_group(0) = ?

+++ exited with 0 +++

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 7

execve("./a.out", ["./a.out", "1000", "7"], 0x7ffe04ffa9e0 /* 30 vars */) = 0

brk(NULL) = 0x55904c6ba000

```
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc8382f9c0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f55d9354000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f55d934d000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f55d9124000
mprotect(0x7f55d914c000, 2023424, PROT_NONE) = 0
mmap(0x7f55d914c000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f55d914c000
mmap(0x7f55d92e1000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f55d92e1000
mmap(0x7f55d933a000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f55d933a000
```

mmap(0x7f55d9340000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f55d9340000

close(3) = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f55d9121000

arch_prctl(ARCH_SET_FS, 0x7f55d9121740) = 0

set_tid_address(0x7f55d9121a10) = 11561

set_robust_list(0x7f55d9121a20, 24) = 0

rseq(0x7f55d91220e0, 0x20, 0, 0x53053053) = 0

mprotect(0x7f55d933a000, 16384, PROT_READ) = 0

mprotect(0x55904aae4000, 4096, PROT_READ) = 0

mprotect(0x7f55d938e000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7f55d934d000, 24583) = 0

getrandom("\x13\x7f\x6f\xfa\xba\xb5\xaa\x56", 8, GRND_NONBLOCK) = 8

brk(NULL) = 0x55904c6ba000

brk(0x55904c6db000) = 0x55904c6db000

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4014800}) = 0

rt_sigaction(SIGRT_1, {sa_handler=0x7f55d91b5870, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO, sa_restorer=0x7f55d9166520}, NULL, 8) = 0

rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f55d8920000

mprotect(0x7f55d8921000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f55d9120910,

parent_tid=0x7f55d9120910, exit_signal=0, stack=0x7f55d8920000, stack_size=0x7fff00, tls=0x7f55d9120640} => {parent_tid=[11562]}, 88) = 11562

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f55d37ff000

mprotect(0x7f55d3800000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f55d3fff910, parent_tid=0x7f55d3fff910, exit_signal=0, stack=0x7f55d37ff000, stack_size=0x7fff00, tls=0x7f55d3fff640} => {parent_tid=[11567]}, 88) = 11567

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

munmap(0x7f55d27fd000, 8392704) = 0

munmap(0x7f55d1ffc000, 8392704) = 0

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=7170100}) = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...}, AT_EMPTY_PATH) = 0

write(1, "The value of the requested threa"..., 38The value of the requested threads: 7

) = 38

write(1, "The total number of threads crea"..., 39The total number of threads created: 8

) = 39

write(1, "Elapsed time: 0.003156 seconds\n", 31Elapsed time: 0.003156 seconds

) = 31

exit_group(0) = ?

+++ exited with 0 +++

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 8

execve("./a.out", ["./a.out", "1000", "8"], 0x7fffd40c4f40 /* 30 vars */) = 0

brk(NULL) = 0x5592764db000

arch_prctl(0x3001 /* ARCH_??? */, 0x7ffcb2ac4090) = -1 EINVAL (Invalid argument)

```
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5e1345e000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f5e13457000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f5e1322e000
mprotect(0x7f5e13256000, 2023424, PROT_NONE) = 0
mmap(0x7f5e13256000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f5e13256000
mmap(0x7f5e133eb000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f5e133eb000
mmap(0x7f5e13444000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f5e13444000
mmap(0x7f5e1344a000, 52816, PROT_READ|PROT_WRITE,
```

MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f5e1344a000

close(3) = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE,

MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5e1322b000

arch_prctl(ARCH_SET_FS, 0x7f5e1322b740) = 0

set_tid_address(0x7f5e1322ba10) = 11573

set_robust_list(0x7f5e1322ba20, 24) = 0

rseq(0x7f5e1322c0e0, 0x20, 0, 0x53053053) = 0

mprotect(0x7f5e13444000, 16384, PROT_READ) = 0

mprotect(0x559274d35000, 4096, PROT_READ) = 0

mprotect(0x7f5e13498000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,

rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7f5e13457000, 24583) = 0

getrandom("\x59\x25\x53\x20\x50\xce\xb6\xf6", 8, GRND_NONBLOCK) = 8

brk(NULL) = 0x5592764db000

brk(0x5592764fc000) = 0x5592764fc000

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5035000}) = 0

rt_sigaction(SIGRT_1, {sa_handler=0x7f5e132bf870, sa_mask=[],

sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,

sa_restorer=0x7f5e13270520}, NULL, 8) = 0

rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE,

MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f5e12a2a000

mprotect(0x7f5e12a2b000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON

E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C

LONE_CHILD_CLEARTID, child_tid=0x7f5e1322a910,

parent_tid=0x7f5e1322a910, exit_signal=0, stack=0x7f5e12a2a000,

stack_size=0x7fff00, tls=0x7f5e1322a640} => {parent_tid=[11574]}, 88) = 11574

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE,

MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f5e12229000

mprotect(0x7f5e1222a000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f5e12a29910,

parent_tid=0x7f5e12a29910, exit_signal=0, stack=0x7f5e12229000,

stack_size=0x7fff00, tls=0x7f5e12a29640} => {parent_tid=[0]}, 88) = 11579

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0

munmap(0x7f5e11a28000, 8392704) = 0

munmap(0x7f5e10a26000, 8392704) = 0

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=6911800}) = 0

newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},

AT_EMPTY_PATH) = 0

write(1, "The value of the requested threa"..., 38The value of the requested threads: 8

) = 38

write(1, "The total number of threads crea"..., 39The total number of threads created: 8

) = 39

write(1, "Elapsed time: 0.001876 seconds\n", 31Elapsed time: 0.001876 seconds

) = 31

exit_group(0) = ?

+++ exited with 0 +++

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 1000 20

execve("./a.out", ["./a.out", "1000", "20"], 0x7fff86d4b0b0 /* 30 vars */) = 0

brk(NULL) = 0x55e005f54000

arch_prctl(0x3001 /* ARCH_??? */, 0x7ffc22ce28e0) = -1 EINVAL (Invalid argument)

mmap(NULL, 8192, PROT_READ|PROT_WRITE,

```
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f95bd828000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f95bd821000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f95bd5f8000
mprotect(0x7f95bd620000, 2023424, PROT_NONE) = 0
mmap(0x7f95bd620000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f95bd620000
mmap(0x7f95bd7b5000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f95bd7b5000
mmap(0x7f95bd80e000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f95bd80e000
mmap(0x7f95bd814000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f95bd814000
```

close(3)   = 0

mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f95bd5f5000

arch_prctl(ARCH_SET_FS, 0x7f95bd5f5740) = 0

set_tid_address(0x7f95bd5f5a10)   = 11621

set_robust_list(0x7f95bd5f5a20, 24)   = 0

rseq(0x7f95bd5f60e0, 0x20, 0, 0x53053053) = 0

mprotect(0x7f95bd80e000, 16384, PROT_READ) = 0

mprotect(0x55e005caf000, 4096, PROT_READ) = 0

mprotect(0x7f95bd862000, 8192, PROT_READ) = 0

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0

munmap(0x7f95bd821000, 24583)   = 0

getrandom("\x7a\x89\x49\x2c\xeb\xeb\xdc\x39", 8, GRND_NONBLOCK) = 8

brk(NULL)   = 0x55e005f54000

brk(0x55e005f75000)   = 0x55e005f75000

clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5015800}) = 0

rt_sigaction(SIGRT_1, {sa_handler=0x7f95bd689870, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO, sa_restorer=0x7f95bd63a520}, NULL, 8) = 0

rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0

mmap(NULL, 8392704, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f95bcdf4000

mprotect(0x7f95bcdf5000, 8388608, PROT_READ|PROT_WRITE) = 0

rt_sigprocmask(SIG_BLOCK, ~[], [], 8)   = 0

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f95bd5f4910, parent_tid=0x7f95bd5f4910, exit_signal=0, stack=0x7f95bcdf4000, stack_size=0x7fff00, tls=0x7f95bd5f4640} => {parent_tid=[11622]}, 88) = 11622

rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f95b77ff000
mprotect(0x7f95b7800000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8)   = 0
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f95b7fff910, parent_tid=0x7f95b7fff910, exit_signal=0, stack=0x7f95b77ff000, stack_size=0x7fff00, tls=0x7f95b7fff640} => {parent_tid=[11629]}, 88) = 11629
rt_sigprocmask(SIG_SETMASK, [], NULL, 8) = 0
futex(0x7f95bd5f4910, FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 11622, NULL, FUTEX_BITSET_MATCH_ANY) = -1 EAGAIN (Resource temporarily unavailable)
munmap(0x7f95ad7fb000, 8392704)   = 0
munmap(0x7f95bc5f3000, 8392704)   = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=13046500}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...}, AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 39The value of the requested threads: 20
) = 39
write(1, "The total number of threads crea"..., 40The total number of threads created: 20
) = 40
write(1, "Elapsed time: 0.008031 seconds\n", 31Elapsed time: 0.008031 seconds
) = 31
exit_group(0)   = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace ./a.out 15000 1
execve("./a.out", ["./a.out", "15000", "1"], 0x7ffc0ee4a3b0 /* 30 vars */) = 0
brk(NULL)   = 0x564a9ae3b000

```
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffda64093d0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5f98196000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f5f9818f000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f5f97f66000
mprotect(0x7f5f97f8e000, 2023424, PROT_NONE) = 0
mmap(0x7f5f97f8e000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f5f97f8e000
mmap(0x7f5f98123000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f5f98123000
mmap(0x7f5f9817c000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f5f9817c000
```

```
mmap(0x7f5f98182000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f5f98182000
close(3)                    = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5f97f63000
arch_prctl(ARCH_SET_FS, 0x7f5f97f63740) = 0
set_tid_address(0x7f5f97f63a10)      = 11701
set_robust_list(0x7f5f97f63a20, 24)    = 0
rseq(0x7f5f97f640e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f5f9817c000, 16384, PROT_READ) = 0
mprotect(0x564a9a486000, 4096, PROT_READ) = 0
mprotect(0x7f5f981d0000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f5f9818f000, 24583)       = 0
getrandom("\xba\x0f\x53\x57\xe1\x18\x89\xfa", 8, GRND_NONBLOCK) = 8
brk(NULL)                   = 0x564a9ae3b000
brk(0x564a9ae5c000)              = 0x564a9ae5c000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5212000}) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5438200}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x6), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 1
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 2
) = 39
write(1, "Elapsed time: 0.000226 seconds\n", 31Elapsed time: 0.000226 seconds
) = 31
exit_group(0)                 = ?
+++ exited with 0 +++
```

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$

Вывод

В ходе выполнения данной лабораторной работы я научилась создавать программы на языке С, обрабатывающие данные в многопоточном режиме, применять стандартные средства операционной системы для управления потоками и их синхронизации.

В ходе тестирования программы выполняющей многопоточную сортировку слиянием, я обнаружила, что увеличение числа потоков приводит к заметному ускорению выполнения программы при обработке больших массивов данных. Недостаток эффективности происходит из-за использования malloc для выделение памяти для подмассивов и частого вызова mutex. Однако эффективность использования вычислительных ресурсов начинает снижаться при превышении количества потоков, соответствующего числу логических ядер процессора. Это связано с дополнительными затратами на управление потоками и синхронизацию между ними.